

8.4.34

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Question

Let (x, y) , be any point on the parabola $y^2 = 4x$. Let **P** be the point that divides the line segment from $(0, 0)$ to (x, y) in the ration 1:3. Then the locus of **P** is.

- ① $x^2 = y$
- ② $y^2 = 2x$
- ③ $y^2 = x$
- ④ $x^2 = 2y$

Theoretical Solution

Given points can be taken to be

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ and } \mathbf{Q} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

Point $\mathbf{P} = \begin{pmatrix} x_p \\ y_p \end{pmatrix}$ divides the line segment $\mathbf{P} - \mathbf{O}$ in the ratio 1:3. so

$$\mathbf{P} = \frac{3\mathbf{O} + \mathbf{Q}}{3 + 1} \quad (2)$$

$$\Rightarrow \begin{pmatrix} x_p \\ y_p \end{pmatrix} = \frac{1}{4} \begin{pmatrix} x \\ y \end{pmatrix} \quad (3)$$

Theoretical Solution

So,

$$x = 4x_p \quad \text{and} \quad y = 4y_p \quad (4)$$

Substituting the above in the equation in the parabola $y^2 = 4x$.

$$y_p^2 = x_p \quad (5)$$

Therefore, the locus of the point **P** is the parabola $y^2 = x$.

```
#include <stdio.h>

void compute_parabola(double *y, double *x, int n) {
    for (int i = 0; i < n; i++) {
        x[i] = (y[i] * y[i]) / 4.0;
    }
}
```

Python Code - Using Shared Object

```
import numpy as np
import matplotlib.pyplot as plt
import ctypes

c_lib=ctypes.CDLL("./code.so")

c_lib.compute_parabola.argtypes = [
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_double),
    ctypes.c_int
]

y = np.linspace(-5,5,100)
x = np.zeros_like(y)
```

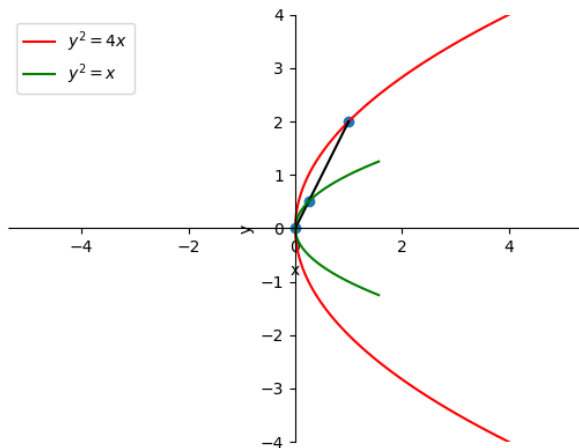
Python Code - Using Shared Object

```
c_lib.compute_parabola(  
    y.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),  
    x.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),  
    y.size  
)  
  
plt.plot(x, y, color = 'red' ,label = "$y^2 = 4x$")  
  
x_p = (3*x + x)/(3+1)  
y_p = (3*y + y)/(3+1)  
  
plt.plot([0,1], [0,2], color = 'black')  
plt.scatter([0,1,0.25], [0,2,0.5])  
  
plt.plot(x_p, y_p, color = 'green', label = "$y^2 = x$")
```

Python Code - Using Shared Object

```
ax = plt.gca()
ax.spines['top'].set_color('none')
ax.spines['bottom'].set_position('zero')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
plt.xlabel('x')
plt.ylabel('y')
plt.legend(loc='best')
plt.grid(False)
plt.axis('equal')
plt.xlim(-2, 2)
plt.ylim(-4, 4)
```


Plot-Using Both C and Python



Python Code

```
import numpy as np
import matplotlib.pyplot as plt
import math

y = np.linspace(-8, 8, 1000)

x = (y*y)/4

plt.plot(x, y, label = "$y^2 = 4x$")

x_p = (3*0 + x)/(3+1)
y_p = (3*0 + y)/(3+1)

plt.plot([0,1], [0,2])
plt.scatter([0,1,0.25], [0,2,0.5])

plt.plot(x_p, y_p, label = "$y^2 = x$")
```

Python Code

```
ax = plt.gca()
ax.spines['top'].set_color('none')
ax.spines['bottom'].set_position('zero')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
plt.xlabel('x')
plt.ylabel('y')
plt.legend(loc='best')
plt.grid(False)
plt.axis('equal')
plt.xlim(-2, 2)
plt.ylim(-4, 4)

plt.savefig("../Figs/plot(py).png")
plt.show()
```

Plot-Using Python only

